

PROJECT REPORT

Bengali Natural Language Processing

Toxicity Detection and Multi-Label Text Classification

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GitHub Repository
arghoghosh88/Natural-Language-Processing-Project

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Abstract

This project focuses on Bengali Natural Language Processing (NLP) for toxicity detection and multi-label text classification. The repository contains datasets, notebooks, generated outputs, a selected best notebook, and a final output file. The work explores machine learning and deep learning approaches for identifying toxic content in Bengali text and predicting toxicity-related categories. The project is organized to support experimentation, evaluation, and submission-oriented model development.

1. Introduction

Online Bengali text contains a wide range of expressions, including neutral, offensive, sarcastic, profane, and otherwise harmful content. Automated toxicity detection can help moderation systems identify potentially harmful text at scale. Bengali presents additional challenges because of linguistic variation, spelling variation, informal writing, code-mixing, and limited high-quality resources compared with English.

2. Problem Statement

The objective is to develop NLP models that can classify Bengali text according to toxicity. The project considers two major settings: binary classification for toxic versus non-toxic content, and multi-label classification for multiple toxicity-related attributes.

3. Objectives

- Build an NLP pipeline for Bengali toxicity classification.
- Perform appropriate text preprocessing and preparation for model training.
- Experiment with machine learning, deep learning, and transformer-based approaches.
- Evaluate models using suitable classification metrics.
- Generate prediction outputs suitable for evaluation or competition submission.
- Maintain an organized repository containing datasets, notebooks, outputs, and selected model code.

4. Repository and Project Organization

The current GitHub repository contains the folders Dataset, Notebooks, and Outputs, along with Best-Notebook.py, Best-output.csv, .gitignore, and README.md.

Component	Purpose
Dataset/	Dataset files used by the project.
Notebooks/	NLP experiments, preprocessing, training, and evaluation notebooks.
Outputs/	Generated model outputs and prediction files.
Best-Notebook.py	Selected/best notebook converted to Python script format.
Best-output.csv	Selected model output/prediction file.
README.md	Project overview and repository documentation.

5. Dataset

The project uses Bengali text data for toxicity classification. The dataset is used for training, validation, testing, and generation of prediction outputs. Exact dataset size, label distribution, and split ratios should be reported from the final dataset version used for the submitted experiment.

6. Data Preprocessing

The preprocessing stage prepares Bengali text for model training. Depending on the selected experiment, preprocessing may include text normalization, cleaning, handling of missing values, tokenization, and conversion of text into model-compatible representations. The exact preprocessing steps should be documented alongside each final experiment to ensure reproducibility.

7. Methodology

The project follows an experimental NLP workflow: dataset preparation, preprocessing, model selection, training, validation, prediction, and evaluation. Multiple experiments are maintained in the Notebooks directory, while the selected implementation is represented by Best-Notebook.py and its associated output.

1. Data loading and inspection
2. Text preprocessing and tokenization
3. Feature or representation generation
4. Model training
5. Validation and hyperparameter experimentation
6. Prediction on evaluation/test data
7. Metric-based evaluation and output generation

8. Models and Experiments

The project explores different NLP modeling approaches, including traditional machine learning, deep learning, and transformer-based methods. The exact model names and configurations should be filled in from the final notebooks before academic submission.

9. Evaluation Metrics

Classification performance can be measured using accuracy, precision, recall, and F1-score. For multi-label classification, label-wise and aggregate metrics can be reported according to the evaluation protocol of the task.

10. Results

The repository currently contains a selected output file, Best-output.csv. Final numerical results should be inserted here from the completed experiment so that no unverified performance value is reported.

Model	Accuracy	F1-score	Notes
[Insert model name]	[Insert]	[Insert]	[Insert]
[Insert model name]	[Insert]	[Insert]	[Insert]
[Insert model name]	[Insert]	[Insert]	[Insert]

11. Error Analysis

Potential sources of error include spelling variations, informal Bengali, mixed Bengali-English text, ambiguous context, sarcasm, short text, and class imbalance. A final error analysis should inspect representative false positives and false negatives and relate them to the model's behavior.

12. Limitations

- Bengali toxicity datasets may contain linguistic and domain-specific biases.
- Informal spelling and code-mixed text can make classification difficult.
- Multi-label examples may contain overlapping or ambiguous categories.
- Model performance depends strongly on the quality and distribution of the available data.

13. Future Work

- Expand and diversify Bengali toxicity datasets.
- Improve preprocessing for spelling variation and code-mixed text.
- Experiment with additional Bengali pretrained language models.
- Perform systematic hyperparameter tuning.
- Use more detailed error analysis and class-wise evaluation.
- Improve reproducibility by documenting exact versions, configurations, and random seeds.

14. Conclusion

This project presents an organized Bengali NLP workflow for toxicity detection and multi-label classification. The repository structure supports experimentation through notebooks and preserves selected code and prediction outputs. The project can be extended through additional data, stronger Bengali language models, improved preprocessing, and more systematic evaluation.

15. GitHub Repository

Repository: <https://github.com/arghosh88/Natural-Language-Processing-Project>

16. References

1. Relevant Bengali toxicity detection dataset/task documentation.
2. Scikit-learn documentation for machine learning and evaluation.
3. PyTorch documentation for deep learning.
4. Hugging Face Transformers documentation for transformer-based NLP.